Prabhavit Gupta

Middle Python Developer (ML): RAG, Hybrid Search & LLM Systems

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (Remote EU / Full Work Authorization)

Summary

Middle Python Developer with 3 years of commercial software engineering experience specializing in production LLM systems, RAG architectures, and information structuring. Hands-on expertise building hybrid search retrieval pipelines combining dense vector embeddings with keyword indexing, context-aware chunking, and metadata enrichment. Proven track record designing offline evaluation harnesses, optimizing prompt-model performance, and structuring unstructured technical documentation into validated relational and entity-linked schemas. Self-driven engineer experienced in owning features end-to-end—from ingestion architecture and model experimentation through automated CI/CD and production deployment.

Experience

FlowSpace.co • *Lead Developer & AI Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Designed and implemented end-to-end Python retrieval and ingestion pipelines, processing unstructured technical documentation and booking records for 4,500+ active enterprise users.
- Built hybrid search architecture integrating dense vector embeddings (OpenAI, Gemini) with PostgreSQL full-text search, tuning chunking strategies and metadata filters to optimize retrieval relevance.
- Implemented schema validation and confidence scoring on LLM outputs, preventing hallucinated entities from reaching downstream services and databases.
- Engineered offline evaluation benchmarks with golden query-response sets, tracking retrieval precision, token expenditure, and latency across prompt and embedding iterations.
- Developed asynchronous job queues and retry handlers with idempotent transaction keys to process large multi-document batches reliably.
- Automated continuous integration and testing using GitHub Actions and Docker, maintaining high test coverage across core ingestion and search APIs.
- *Tools: Python, RAG, Vector Embeddings, Hybrid Search, PostgreSQL, Supabase, Docker, Linux*

Urlink • *Project Manager & Solutions Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Built Python data transformation and entity extraction scripts connecting client documentation streams to structured PostgreSQL databases.
- Automated API verification pipelines and schema validation routines, ensuring data integrity across interconnected business platforms.
- Managed technical project execution end-to-end: gathered functional specifications, mapped entity relationships, and produced clear technical documentation.
- *Tools: Python, REST APIs, PostgreSQL, Supabase, TypeScript*

Independent Technical Consultant • *AI Automation & Integration Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Developed automated scraping and ingestion pipelines in Python, extracting structured domain data and technical catalog content from web sources.
- Created custom evaluation scripts and webhook monitors to detect payload drift and schema mismatches between API endpoints.
- Documented data schemas, entity models, and API integration runbooks for engineering stakeholders.
- *Tools: Python, Supabase, Webhooks, REST APIs, Linux*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Built automated workflows integrating LLM endpoints for text processing, verification flows, and user notifications.
- *Tools: Python, OpenAI API, Sendgrid, REST APIs*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** European Long Term Residence Permit (Italy), unrestricted EU work authorization (Partita IVA / B2B ready). No sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (beginner).
- **Availability:** Available immediately for full-time remote roles across European time zones.

Core Skills

- **Python & AI Engineering:** Python, LLMs (OpenAI, Anthropic Claude, Google Gemini), RAG architecture, vector embeddings, chunking strategies, metadata enrichment, prompt design.
- **Retrieval & Search:** Hybrid search (dense vector retrieval + sparse keyword search), relevance tuning, reranking, semantic filtering, knowledge structuring.
- **Evaluation & Quality:** Offline eval harnesses, golden dataset benchmarking, regression test suites, token budget optimization, hallucination prevention.
- **Data & Backend Systems:** PostgreSQL, Supabase, RESTful APIs, relational data modeling, asynchronous job queues, Docker containerization, Linux environments.
- **Developer Workflows & Reliability:** Git, GitHub Actions CI/CD, unit/integration testing, automated data validation pipelines, technical documentation.